

EY Physical AI Engineering | Independent candidate materials

Russell Dudek

Physical AI Delivery & Operations Leader | Robotics, Computer Vision & High-Reliability Systems

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<https://russelldudek.github.io/EY/>

Role-aligned profile

Operator-technologist who connects physical systems, engineering evidence, client operations and human authority. Current leader in a software-enabled robotics and machine-vision environment; background spans regulated autonomy, high-reliability electronics, scaled Amazon operations, AI workflow design and cross-functional technical programs.

Core capabilities

Physical AI delivery Robotics operations Computer-vision concepts High-reliability engineering
Technical workstreams Client-operating translation AI readiness & governance Lean / Gemba / Six Sigma
Quality systems Cross-functional change

Professional experience

Vape-Jet | Director of Operations

Leads AI-first operating transformation across production, purchasing, quality, customer success, support, engineering issue flow, Odoo/ERP, Jira, HubSpot, Slack, telemetry and AI-assisted workflows.

Connects factory, field, customer and machine-fleet signals into prioritized work, earlier risk detection, cleaner handoffs, dashboards, standard work, training and accountability.

Works directly in a software-enabled robotics and machine-vision environment with approximately 1,000 fielded robots informing operating and support feedback loops.

DudeWorth | Founder / AI and Automation Advisor

Develops AI-readiness and opportunity frameworks assessing business value, feasibility, data readiness, governance, adoption burden, reuse and time-to-value.

Designs agentic workflow patterns using structured context, retrieval, AI agents, human judgment, escalation, decision records, evaluation and governance rails.

Advises on AI-enabled operations, warehouse automation, AMRs, ASRS, cobots, transportation systems, network analysis and workflow design.

ZeusVu | Chief Pilot, Autonomous Systems & Aerial Data Operations

Led regulated drone operations across medical, energy, logistics, real-estate and restricted-airspace contexts with a 100% safety record and no FAA incidents.

Developed TensorFlow-based concepts for automated aerial inventory and computer-vision-enabled field intelligence.

Designed medical-drone delivery and chain-of-custody concepts connecting technology, risk, stakeholders and governed execution.

Amazon | Operations Management / Site Leader

Led launch and execution for Amazon Prime Pittsburgh in a 24/7 customer-backward environment supporting approximately five million second-day orders annually.

Built leadership routines, labor planning, throughput visibility, escalation paths, standard work and operating mechanisms across safety, quality and customer experience.

Contributed to logistics innovation through Prime Air research and Amazon Flex geolocation concepts.

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Professional experience, continued

Compunetics | Director of Operations and Engineering Management

Progressed through R&D program management, production management, strategic acquisition, operations management and director-level leadership.

Led engineering, manufacturing, quality, customer delivery and complex technical programs in advanced electronics and high-reliability environments.

Built quality systems, standard work, root-cause practices, performance measures and process controls.

Collaborated with high-consequence customers and partners including DoD, CERN, Intel and AMD; published industry articles, delivered technical talks and participated in the HyperTransport Consortium.

Cardinal Building Products | Regional Director

Built the company's first remote regional operation and modernized distributed customer, CRM, commercial, fulfillment, logistics, digital-marketing and online-buying workflows.

Advanced AI-enhanced transportation-management and route-planning concepts.

Additional leadership

Enviro Social Capital | Chairman of the Board

Co-founded a public-private initiative focused on sustainable infrastructure, green finance and civic transformation; helped shape a \$200 million Social Impact Bond feasibility concept.

IEEE | CMPT and EDS Pittsburgh Chapter Chair

Led regional technology programming connecting industry, academia, STEM outreach, electronics, manufacturing and emerging technology.

Education

University of Pittsburgh | Bachelor of Science

Academic grounding includes materials science, physical chemistry and biology-related studies.

Academic work is indexed through the candidate-provided Google Scholar profile:

scholar.google.com/citations?user=yHQSVD8AAAAJ .

Credentials

Google AI Essentials

IBM Enterprise Design Thinking Practitioner

Six Sigma Certification

OSHA 10

Technical toolkit

Agentic AI | LLM workflows | RAG patterns | Human-in-the-loop design | Python | SQL | JavaScript | TensorFlow | Tableau | ERP/Odoo | Jira | Confluence | HubSpot | Slack | GIS and mapping workflows

Evidence boundary

Does not claim production ownership of ROS, NVIDIA Omniverse, MLflow, HIL/SIL pipelines or named hyperscale ML platforms. The transfer case rests on verified physical-systems, computer-vision, high-reliability, operating and client-delivery evidence.

Role proposition

Close the gap between what the model can do and what the client can trust. Lead the technical workstream through evidence that connects simulation, embodied behavior, human authority, deployment discipline, operating ownership and reusable learning.